01 / PROBLEM AND CONTRIBUTION

Rectifier-fed buck converter prototype

ELE 3261 · Power Electronics Lab | May 2023 | Four-person team

RESULT AT A GLANCE

Connected rectifier and buck-converter analysis with LTspice/Simulink waveforms and hardware observations, including documented no-load and full-load regulation calculations.

The engineering question

How can rectification and switching conversion be combined to produce a lower DC output for a charging application?

Situation

A charging-oriented power-conversion prototype needs rectification, smoothing, and a controlled lower DC output. The project combines circuit calculations, simulation, and hardware observations.

Task

Contribute component calculations, LTspice/Simulink evaluation, hardware work, and documentation as part of the team.

My contribution

Hardware, calculations, simulations, and report work, shared with the project team.

Tools and concepts

Power electronics / LTspice / Simulink / MOSFET switching / Hardware prototyping

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

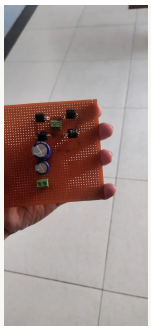
Combined a full-bridge rectifier and smoothing stage with a MOSFET-switched buck converter. Used inductor/capacitor calculations to establish the output filter.

02 / Implementation step

Compared simulated waveforms in LTspice and Simulink with hardware output observations. The report includes the prototype photographs and load-dependent output measurements.

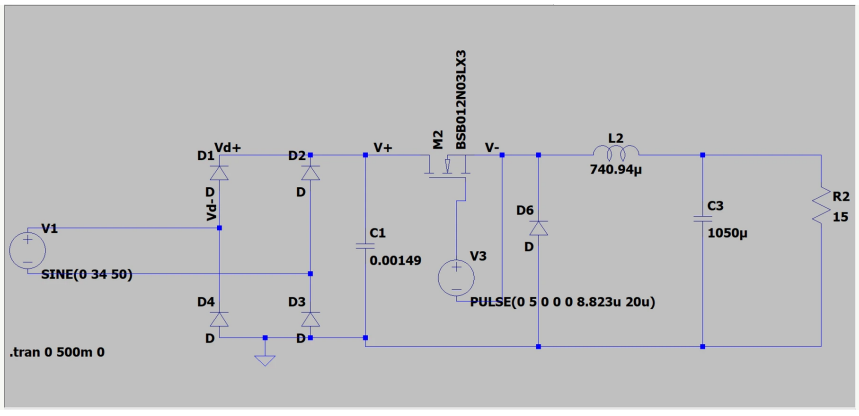
Stages of conversion

The documented path combines rectification and smoothing with a switched buck stage and output filter. Component calculations and waveform observations describe different parts of that path; prototype photographs establish the hardware context.



Rectifier hardware prototype

Full-bridge diode rectifier hardware photographed in the team report.



Rectifier and buck simulation circuit

LTspice schematic from the project report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

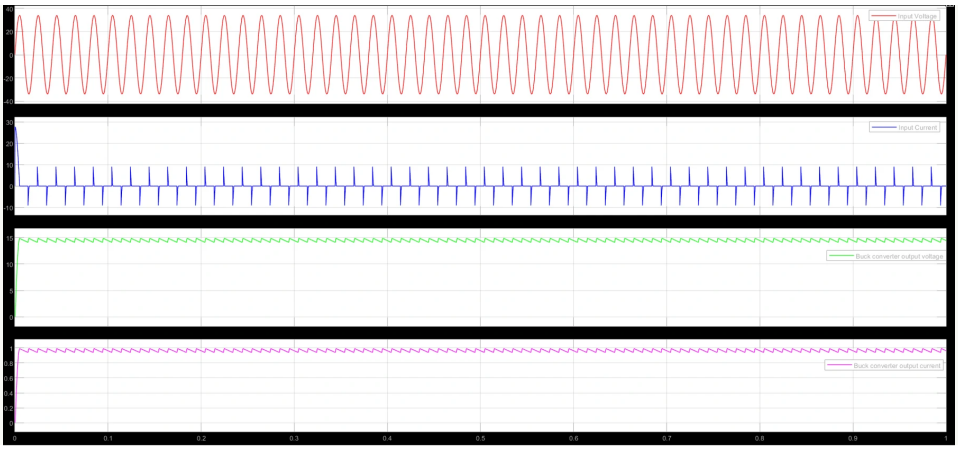
Evidence, interpretation, and next steps

Reported result

The project report documents both simulation and hardware evaluation, including a voltage-regulation calculation from the no-load and full-load values.

Interpreting regulation

No-load and full-load voltages are used for the report's regulation calculation. That comparison is different from efficiency, which would require matched input/output power measurements under specified operating conditions.



Simulated voltage and current waveforms

Simulation results from the rectifier and buck-converter study.

Scope and limitations

No measured charging-efficiency improvement is claimed. This case study focuses on the documented circuit, component calculations, and evaluation.

Possible next steps - proposed, not completed results

- Provide a labelled table linking each hardware measurement to its load and input conditions.
- Measure input/output power before adding any charging-efficiency claim.

Source material: pe_miniproject_report.docx, chapters 2-4; buckconverter.asc, buck_converter.m, and Simulink model in gptz.zip.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: Rectifier-based-buck-converter-prototype-for-charging-applications